Ricardo Semião e Castro

Experience

Research Consultant
ADB Sindical (syndicate of diplomats) *08/2024 - ongoing*
Research on the diplomatic career path and its barriers. Creation of a predictive calculator for individual's careers (atualizamre.com.br/calculadora). Data scrapping, reorganization, and validation from different media sources.

Research Assistant
NYU-Abu Dhabi & FGV-EESP *06/2026 - 09/2026*
The empirical implementation of "Predictive Conditional Alphas" Corradi et. al (2013). Processing of high-frequency stock databases, implementation of custom high-frequency and nonparametric alpha/beta estimators, and creation of a fully documented replication package.

Research Assistant
FGV-EESP & Ripple *10/2022 - 04/2023*
Development of a R-Shiny application for crypto portfolio optimization and analysis.

Econometrics I Teaching Assistant
FGV-EESP *02/2022 - 06/2023*
Conducted exercises, theoretical classes, and R teaching sessions.

Vice President, HR Director
GVCode (data science league at FGV) *06/2020 - 12/2023*
Coordinated all league activities, created and conducted selection processes and trainee programs. Data science projects.

Private Tutor in Statistics, R, and Python
Private lessons for college students. Active in StackOverflow: 260 answers in over 125 tags, peaked at top 5% in the R community.

Projects

- Code packages: predicater - modern validation of R objects; hfpca - PCA for high frequency data; phyopt - numerical optimization with population-based hybrid algorithms.
- Educational material about R (RFCD), with classes being given at GV's teaching assistantship and GVCode.
- Thesis: Masters thesis - performance analysis of regime-switching models via Monte Carlo simulations; Graduation thesis - SVAR models applied to epidemiology.
- My personal website built with bare HTML/SCSS/JS, and python scripts to orchestrate components and the blog.

Contact

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Skills

- **Languages:** Fluent English (CAE)
- **Programming:** R, Python; JS, Julia, Stata
- **Document:** LaTeX, Web, Quarto, Office
- **Tools:** Git/GitHub, SQL; Docker

Education

Bachelor's Degree in Economics - *FGV-EESP* (*2019 - 2023*)
Master's Degree in Economics - *FGV-EESP* (*2024 - 2026*)

Awards

- ANPEC exam, the Brazil-wide admission for master's students in economics.
  10th place (2023).
- Prediction challenge for investment flow in the Brazilian stock exchange, using a neural network.
  2nd place (2022, Constellation A. M.).
- Academic assignments: Ridge and Lasso regression on salary determinants; Nations clustering with WTO data.
  1st and 2nd place in the category (2020 and 2021, FGV-EESP).

Other Activities

- Voluntary work during high school. Worked in daycare, nursing homes, and hospitals.
- Volunteer teacher in the EESP Ensina Project (2022 - 2023). Taught economics to high school students.
- Member of the Scout Movement (2010 - 2021). Developed socio-environmental activities.